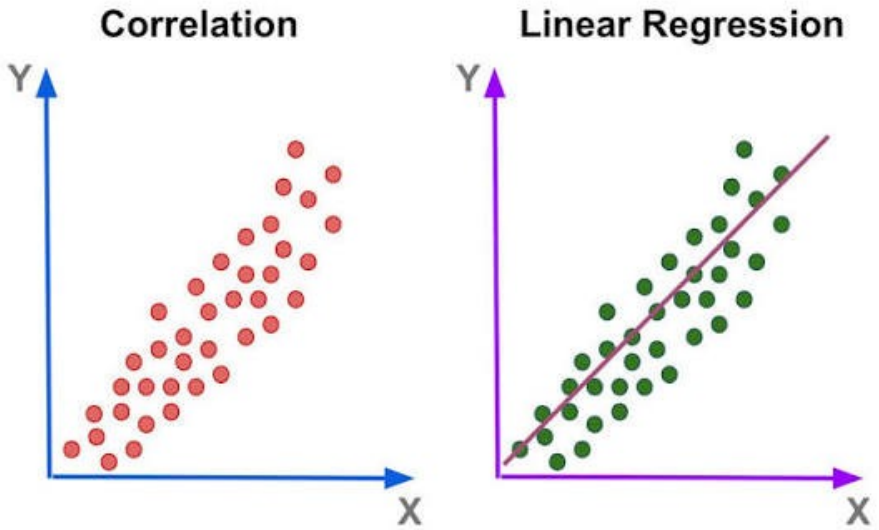
Lecture 27

Prediction III



Announcements

- Course project directed questions are out!

- Office hours every Tuesday (noon to 4pm) and Friday (11am to 1pm)
  - Link to book appointments is on the course website

Today

How to evaluate a regression model?

Delivery Robots

Suppose the dining halls ran a trial using delivery robots on campus

- Tracked delivery time, distance, items damaged, and snow depth
- Trial lasted for 300 deliveries



Using data from the trial, we're asked to

1) Predict how long a delivery will take
2) Predict whether or not items will be damaged

Language of Prediction

Variables

- X denotes inputs X = { distance, snow depth }
- Y denotes output

Goal: predict Y using X

- If Y is numerical, this is called **regression** Y = delivery time
- If Y is categorical, this is called **classification** Y = damaged items

Often call X the **feature space** and individual inputs as **features**

Predictive Model

Rule mapping inputs to outputs that's *learned* from the data

- Give the model some inputs and get an output
- Example: input distance, snow depth and receive delivery time as output

Can use the model without having the access to the data

- Models are a type of statistic

Common model types

- Linear models, trees (e.g., decision trees, random forests), neural networks

Linear Regression

Idea: find the linear function of our inputs that best fits the data

- "best fits" means minimizes squared error between true output and predicted output

Predict Y on using inputs $X_1, X_2, \ldots, X_p$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_p X_p$$

Accessing Model Performance

There are several metrics for accessing model performance

- We'll talk about two: squared error (regression) and accuracy (classification)

Goal: want to know how good predictions are when we predict on new data

Accessing Model Performance

There are several metrics for accessing model performance

- We'll talk about two: mean squared error (regression) and accuracy (classification)

Goal: want to know how good predictions are when we predict on new data

- Why?

Accessing Model Performance

There are several metrics for accessing model performance

- We'll talk about two: mean squared error (regression) and accuracy (classification)

Goal: want to know how good predictions are when we predict on new data

- Why?
- Generalizing to unseen data is what makes a model useful

Mean Squared Error

Average squared difference between the predicted and the actual value

- Used for regression (where the target is a number)
- Nice mathematical properties that makes optimization easy (convex)

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

actual

predicted

Splitting Data



We typically split out dataset into train, validate, and test

Splitting Data

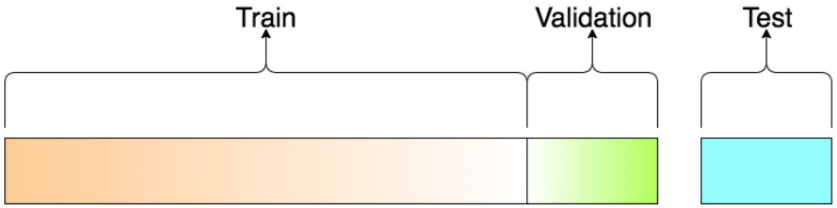


We typically split out dataset into train, validate, and test

Training Data

- Data we use to train a model (e.g. solving for parameters of linear regression)

Splitting Data



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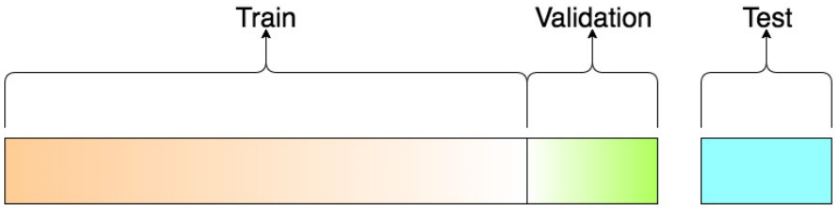
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Training Data

- Data we use to train a model (e.g. solving for parameters of linear regression)

Validation Data

- Data we use for model selection (when we're choosing between multiple models)

Test Data

- Data used to measure performance on unseen data (how well the model generalizes)
- The model never sees this data during training

What is good?

Want: a model that gives high test data performance (low test error)

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Why not settle for a model that has high training data performance?

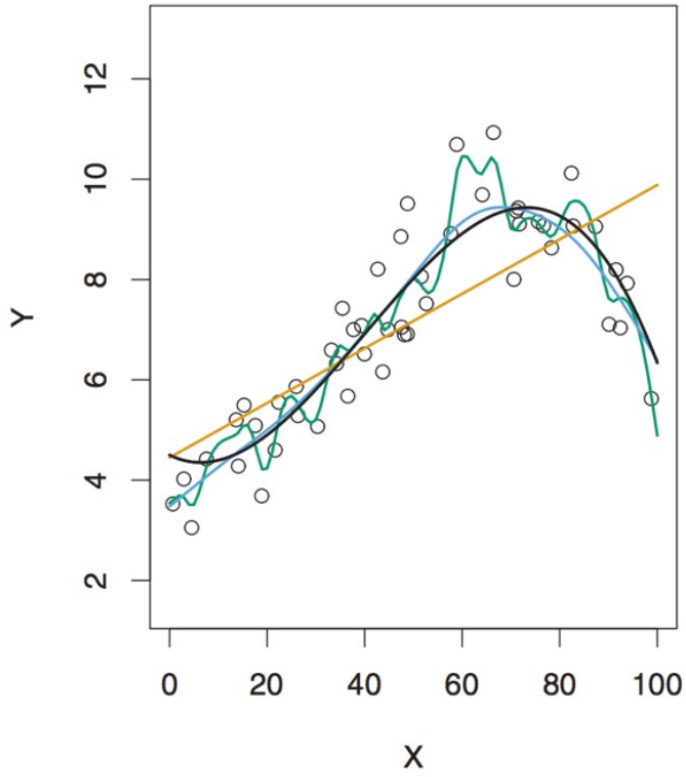
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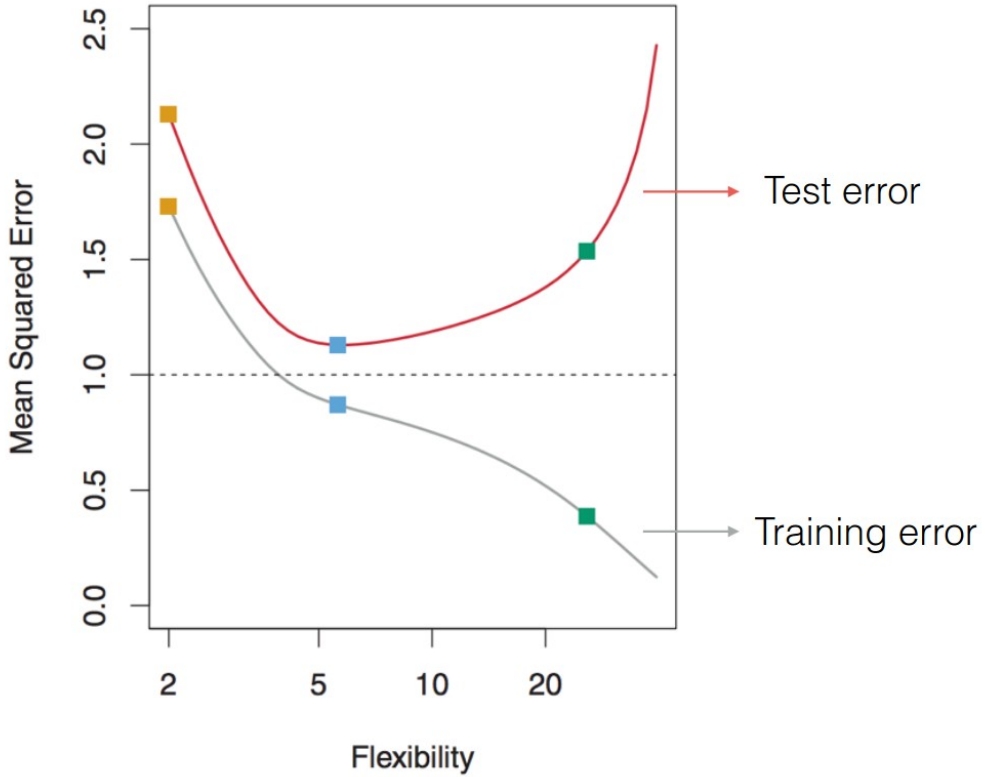
Want: a model that gives high test data performance (low test error)

Why not settle for a model that has high training data performance?

- i.e. low error on the training data
- There's no guarantee that the method with the highest training data performance will have the highest test data performance
- In many cases, it will not (*overfitting*)

Model Flexibility





Model Flexibility

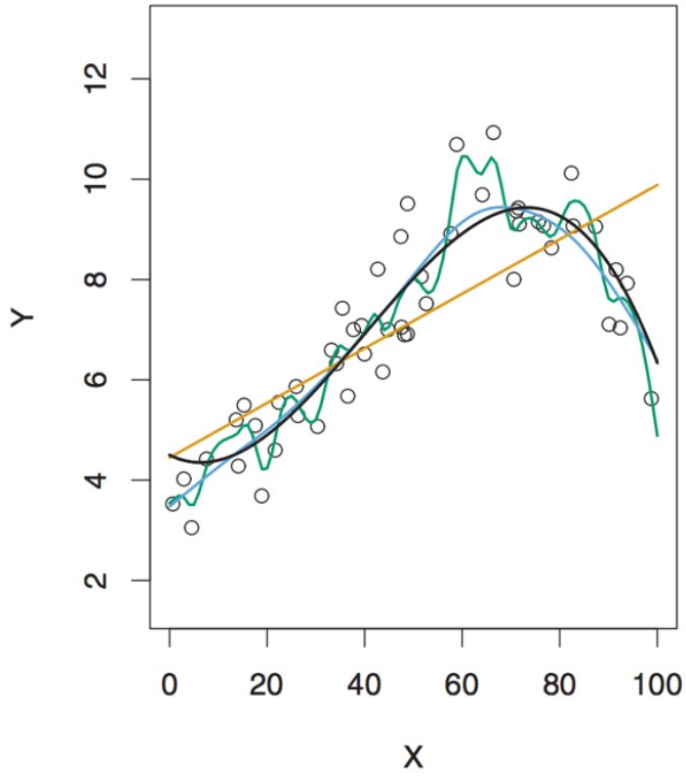
As model flexibility increases, training error decreases

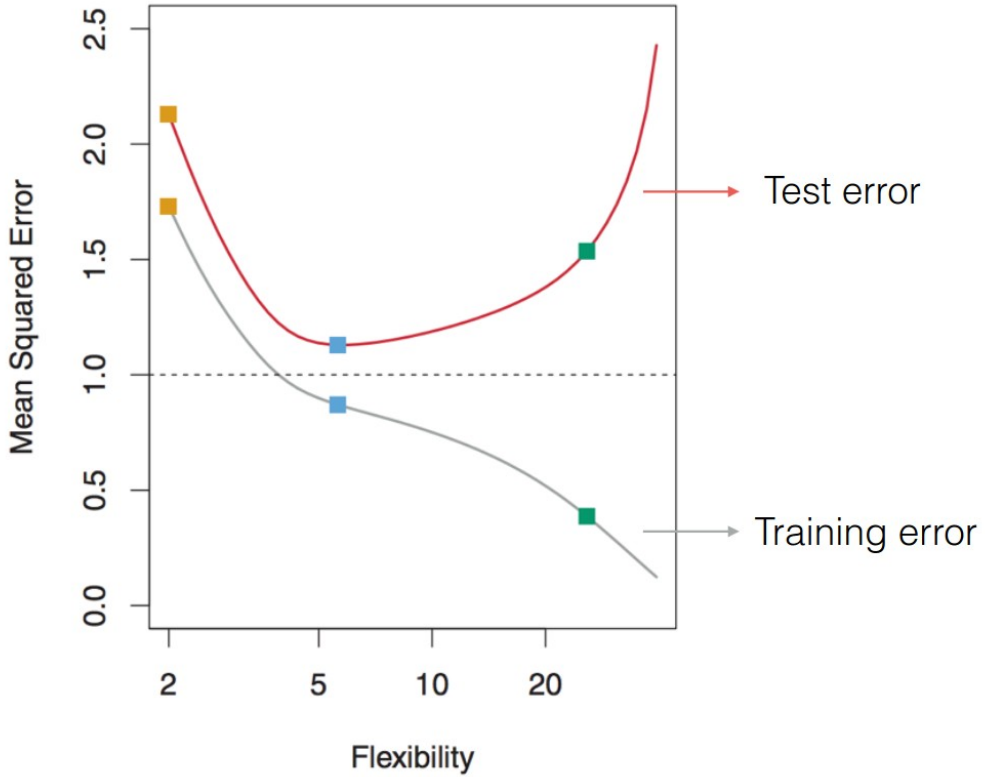
- Flexibility allows a model to better *fit* the training data
- Some of the variability in the training data may be noise, however

The test error may or may not decrease

If the test error is much larger than the training error, then the model is **overfitting**

Model Flexibility

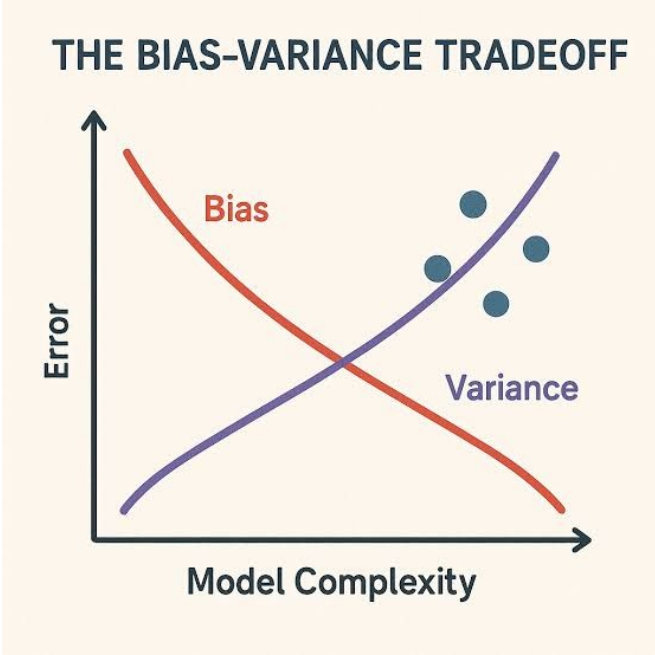




Bias and Variance

Expected test data error = $\texttt{Variance} + \texttt{Bias}^2 + \sigma^2$

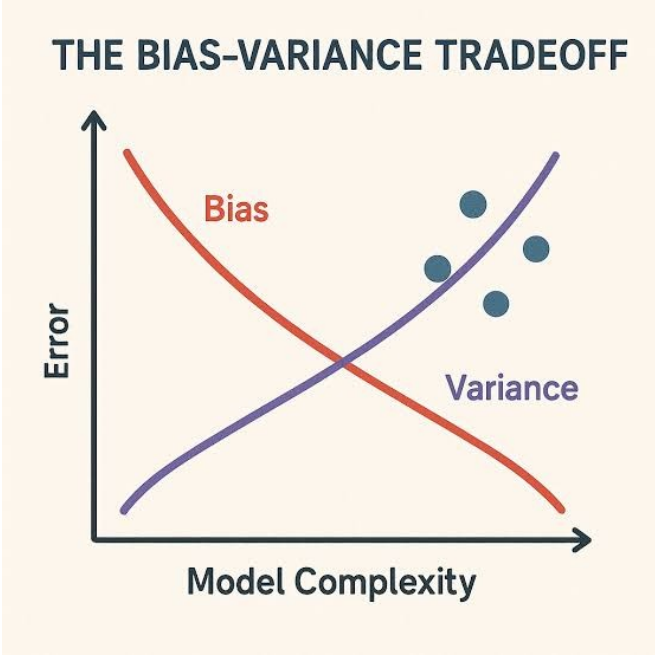
- On average, how does our model perform on new data
- **`Bias`**: error caused by simplifying assumptions built into the model
- **`Variance`**: how much of the learned function will change if trained on different training data



Bias and Variance

Generally, more flexible methods have more variance and less bias (and vice versa)

The **best method** for a prediction task will balance the two types of error to achieve the **lowest test error**



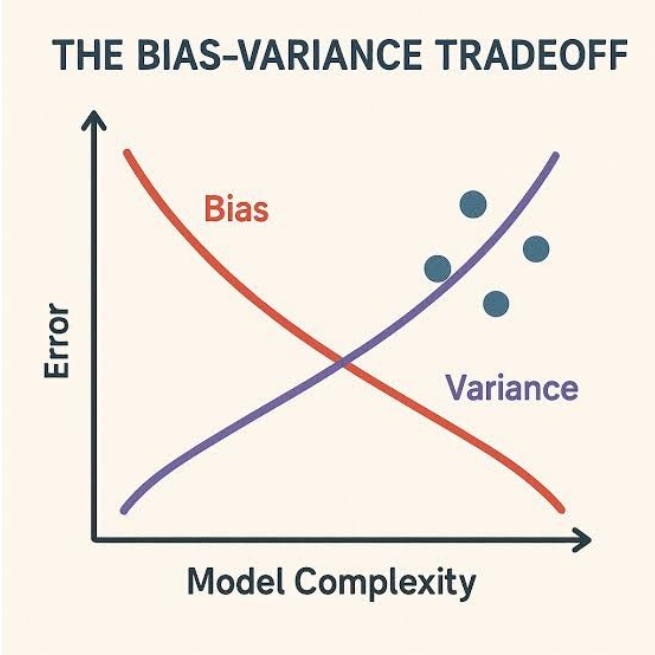
Bias and Variance

Generally, more flexible methods have more variance and less bias (and vice versa)

The **best method** for a prediction task will balance the two types of error to achieve the **lowest test error**

In statistics, we usually *only care* about unbiased models

- Statistics typically tries to learn about the world
- In ML, we're just trying to predict well



Going back to linear regression

Nice properties

- Simple model
- Solves for coefficients that minimize MSE on training data
- Interpretable coefficients

Weak properties

- May be too simple to make accurate predictions
- Sensitive to outliers in the training data due to MSE

Next Time

How to evaluate a classification model?